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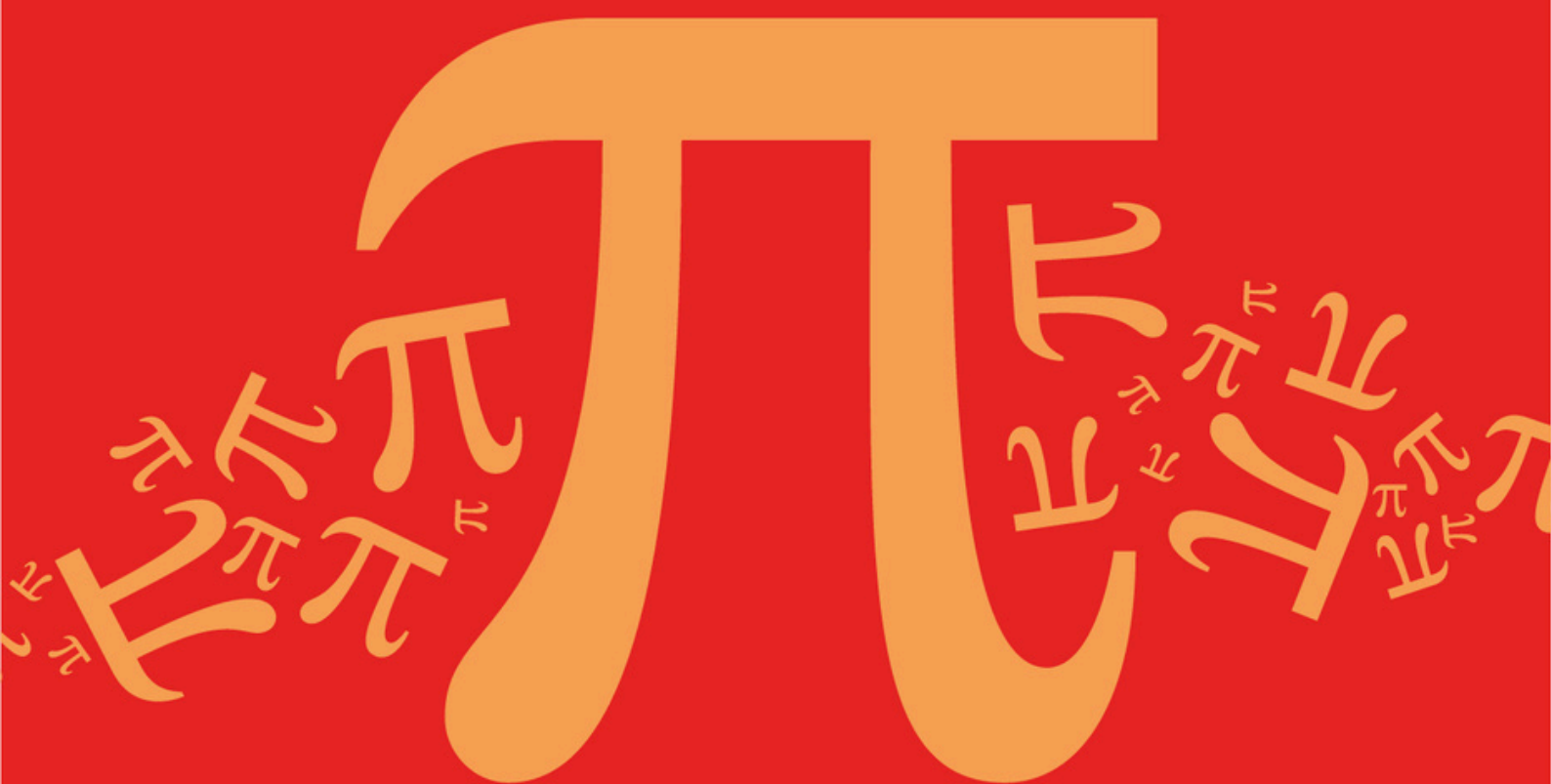
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AQA A Level Further Mathematics (7367)



Topic

C: Matrices

Sub topic

C2: Zero and Identity

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

C: Matrices

Sub topic

C2: Zero and Identity

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

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Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

Two matrices **A** and **B** satisfy the equation

$$\mathbf{AB} = \mathbf{I} + 2\mathbf{A}$$

where **I** is the identity matrix and
$$\mathbf{B} = \begin{bmatrix} 3 & -2 \\ -4 & 8 \end{bmatrix}$$

Find **A**.

(Total 3 marks)

Q2.

(a) The matrix **X** is defined by
$$\begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$$

(i) Given that $\mathbf{X}^2 = \begin{bmatrix} m & 2 \\ 3 & 6 \end{bmatrix}$, find the value of m .

(1)

(ii) Show that $\mathbf{X}^3 - 7\mathbf{X} = n\mathbf{I}$, where n is an integer and **I** is the 2×2 identity matrix.

(4)

(b) It is given that $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.

(i) Describe the geometrical transformation represented by **A**.

(1)

(ii) The matrix **B** represents an anticlockwise rotation through 45° about the origin.

Show that $\mathbf{B} = k \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$, where k is a surd.

(2)

(iii) Find the image of the point $P(-1, 2)$ under an anticlockwise rotation through 45° about the origin, followed by the transformation represented by **A**.

(4)

(Total 12 marks)

Q3.

The matrix **A** is given by

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$$

- (a) Given that $\mathbf{A}^2 = \begin{bmatrix} p & -2 & -4 \\ 5 & 6 & 4 \\ 10 & q & 9 \end{bmatrix}$, find the value of p and the value of q . (2)

- (b) Given that $\mathbf{A}^3 - 6\mathbf{A}^2 + 11\mathbf{A} - 6\mathbf{I} = \mathbf{0}$, prove that

$$\mathbf{A}^{-1} = \frac{1}{6}(\mathbf{A}^2 - 6\mathbf{A} + 11\mathbf{I}) \quad (2)$$

- (c) Given that $\mathbf{A}^{-1} = \frac{1}{6} \begin{bmatrix} r & -2 & 2 \\ -1 & 5 & -2 \\ -2 & s & 2 \end{bmatrix}$, find the value of r and the value of s . (2)

- (d) Hence, or otherwise, find the solution of the system of equations

$$\begin{aligned} x - z &= k \\ x + 2y + z &= 5 \\ 2x + 2y + 3z &= 7 \end{aligned}$$

giving your answers in terms of k .

(3)

(Total 9 marks)

Q4.

The matrices $\mathbf{A}$ and $\mathbf{B}$ are defined by

$$\mathbf{A} = \begin{bmatrix} p & 2 \\ 4 & p \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix}$$

- (a) Find, in terms of p , the matrices:

(i) $\mathbf{A} - \mathbf{B}$; (1)

(ii) $\mathbf{AB}$. (2)

- (b) Show that there is a value of p for which $\mathbf{A} - \mathbf{B} + \mathbf{AB} = k\mathbf{I}$, where k is an integer and $\mathbf{I}$ is the 2×2 identity matrix, and state the corresponding value of k .

(4)

(Total 7 marks)

Q5.

- (a) Using surd forms, find the matrix of a rotation about the origin through 135° anticlockwise.

(2)

(b) The matrix $\mathbf{M}$ is defined by $\mathbf{M} = \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix}$.

- (i) Given that $\mathbf{M}$ represents an enlargement followed by a rotation, find the scale factor of the enlargement and the angle of the rotation.

(3)

- (ii) The matrix $\mathbf{M}^2$ also represents an enlargement followed by a rotation. State the scale factor of the enlargement and the angle of the rotation.

(2)

- (iii) Show that $\mathbf{M}^4 = k\mathbf{I}$, where k is an integer and $\mathbf{I}$ is the 2×2 identity matrix.

(2)

- (iv) Deduce that $\mathbf{M}^{2012} = -2^n \mathbf{I}$ for some positive integer n .

(2)

(Total 11 marks)

Q6.

The matrix $\mathbf{A} = \begin{bmatrix} k & 1 & 2 \\ 2 & k & 1 \\ 1 & 2 & k \end{bmatrix}$, where k is a real constant.

- (a) (i) Show that there is a value of k for which

$$\mathbf{A} \mathbf{A}^T = m \mathbf{I}$$

where m is a rational number to be determined and $\mathbf{I}$ is the 3×3 identity matrix.

(6)

- (ii) Deduce the inverse matrix, $\mathbf{A}^{-1}$, of $\mathbf{A}$ for this value of k .

(1)

- (b) (i) Find $\det \mathbf{A}$ in terms of k .

(2)

- (ii) In the case when $\mathbf{A}$ is singular, find the integer value of k and show that there are no other possible real values of k .

(3)

- (iii) Find the value of k for which $\lambda = 7$ is a real eigenvalue of $\mathbf{A}$.

(2)

(Total 14 marks)

Q7.

$$\mathbf{X} = \begin{bmatrix} 3 & 1 & -3 \\ 2 & 4 & 3 \\ -4 & 2 & -1 \end{bmatrix}.$$

The non-singular matrix

- (a) (i) Show that $\mathbf{X}^2 - \mathbf{X} = k\mathbf{I}$ for some integer k . (3)

- (ii) Hence show that $\mathbf{X}^{-1} = \frac{1}{20}(\mathbf{X} - \mathbf{I})$. (2)

$$\mathbf{Y}^{-1} = \begin{bmatrix} 60 & 0 & 0 \\ 0 & 0 & -10 \\ 0 & 20 & 0 \end{bmatrix}.$$

- (b) The 3×3 matrix $\mathbf{Y}$ has inverse

Without finding $\mathbf{Y}$, determine the matrix $(\mathbf{XY})^{-1}$.

(3)
(Total 8 marks)

Q8.

The matrix $\mathbf{A}$ is defined by

$$\mathbf{A} = \begin{bmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{bmatrix}$$

- (a) (i) Calculate the matrix $\mathbf{A}^2$ (2)

- (ii) Show that $\mathbf{A}^3 = k\mathbf{I}$, where k is an integer and $\mathbf{I}$ is the 2×2 identity matrix. (2)

- (b) Describe the single geometrical transformation, or combination of two geometrical transformations, corresponding to each of the matrices:

- (i) $\mathbf{A}^3$; (2)

- (ii) $\mathbf{A}$. (3)

(Total 9 marks)

Q9.

It is given that

$$\mathbf{A} = \begin{bmatrix} 1 & 4 \\ 3 & 1 \end{bmatrix}$$

and that $\mathbf{I}$ is the 2×2 identity matrix.

- (a) Show that $(\mathbf{A} - \mathbf{I})^2 = k\mathbf{I}$ for some integer k .

(3)

- (b) Given further that

$$\mathbf{B} = \begin{bmatrix} 1 & 3 \\ p & 1 \end{bmatrix}$$

find the integer p such that

$$(\mathbf{A} - \mathbf{B})^2 = (\mathbf{A} - \mathbf{I})^2$$

(4)

(Total 7 marks)

Q10.

The matrices $\mathbf{A}$ and $\mathbf{B}$ are defined in terms of a real parameter t by

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 1 \\ 2 & t & 4 \\ 3 & 2 & -1 \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} 15 & -4 & -1 \\ -2t & 4 & 2 \\ 17 & -4 & -3 \end{bmatrix}$$

- (a) Find, in terms of t , the matrix $\mathbf{AB}$ and deduce that there exists a value of t such that $\mathbf{AB}$ is a scalar multiple of the 3×3 identity matrix $\mathbf{I}$.

(5)

- (b) For this value of t , deduce $\mathbf{A}^{-1}$.

(2)

(Total 7 marks)

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Q11.

The matrices $\mathbf{A}$ and $\mathbf{B}$ are defined by

$$\mathbf{A} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 1 & 1 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & 1 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

Describe fully the geometrical transformation represented by each of the following matrices:

- (a) $\mathbf{A}$;

(2)

- (b) $\mathbf{B}$;

(2)

- (c) $\mathbf{A}^2$;

(2)

- (d) $\mathbf{B}^2$;

(2)

(e) $\mathbf{AB}$.

(3)

(Total 11 marks)

Q12.

The matrices $\mathbf{A}$ and $\mathbf{B}$ are defined by

$$\mathbf{A} = \begin{bmatrix} k & k \\ k & -k \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -k & k \\ k & k \end{bmatrix}$$

where k is a constant.

(a) Find, in terms of k :

(i) $\mathbf{A} + \mathbf{B}$;

(1)

(ii) $\mathbf{A}^2$.

(2)

(b) Show that $(\mathbf{A} + \mathbf{B})^2 = \mathbf{A}^2 + \mathbf{B}^2$.

(4)

(c) It is now given that $k = 1$.

(i) Describe the geometrical transformation represented by the matrix $\mathbf{A}^2$.

(2)

(ii) The matrix $\mathbf{A}$ represents a combination of an enlargement and a reflection. Find the scale factor of the enlargement and the equation of the mirror line of the reflection.

(3)

(Total 12 marks)

Q13.

The matrix $\mathbf{M}$ is defined by

$$\mathbf{M} = \begin{bmatrix} \sqrt{3} & 3 \\ 3 & -\sqrt{3} \end{bmatrix}$$

(a) (i) Show that

$$\mathbf{M}^2 = p\mathbf{I}$$

where p is an integer and $\mathbf{I}$ is the 2×2 identity matrix.

(3)

(ii) Show that the matrix $\mathbf{M}$ can be written in the form

$$q \begin{bmatrix} \cos 60^\circ & \sin 60^\circ \\ \sin 60^\circ & -\cos 60^\circ \end{bmatrix}$$

where q is a real number. Give the value of q in surd form.

(3)

- (b) The matrix $\mathbf{M}$ represents a combination of an enlargement and a reflection.

Find:

- (i) the scale factor of the enlargement;

(1)

- (ii) the equation of the mirror line of the reflection.

(1)

- (c) Describe fully the geometrical transformation represented by $\mathbf{M}^4$.

(2)

(Total 10 marks)

Q14.

The non-singular matrix $\mathbf{M} = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & -1 & 2 \end{bmatrix}$.

- (a) (i) Show that

$$\mathbf{M}^2 + 2\mathbf{I} = k\mathbf{M}$$

for some integer k to be determined.

(3)

- (ii) By multiplying the equation in part (a)(i) by $\mathbf{M}^{-1}$, show that

$$\mathbf{M}^{-1} = a\mathbf{M} + b\mathbf{I}$$

for constants a and b to be found.

(3)

- (b) (i) Determine the characteristic equation of $\mathbf{M}$ and show that $\mathbf{M}$ has a repeated eigenvalue, 1, and another eigenvalue, 2.

(6)

- (ii) Give a full set of eigenvectors for each of these eigenvalues.

(5)

- (iii) State the geometrical significance of each set of eigenvectors in relation to the transformation with matrix $\mathbf{M}$.

(3)

(Total 20 marks)

Q15.

The matrices **A** and **B** are given by

$$\mathbf{A} = \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$$

- (a) Calculate the matrix **AB**. (2)
- (b) Show that **A**² is of the form $k\mathbf{I}$, where k is an integer and **I** is the 2×2 identity matrix. (2)
- (c) Show that $(\mathbf{AB})^2 \neq \mathbf{A}^2\mathbf{B}^2$. (3)

(Total 7 marks)

Q16.

The matrix $\mathbf{P} = \begin{bmatrix} 4 & -1 & 2 \\ 1 & 1 & 3 \\ -2 & 0 & a \end{bmatrix}$, where a is constant.

- (a) (i) Determine $\det \mathbf{P}$ as a linear expression in a . (2)
- (ii) Evaluate $\det \mathbf{P}$ in the case when $a = 3$. (1)
- (iii) Find the value of a for which **P** is singular. (2)

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- (b) The 3×3 matrix **Q** is such that $\mathbf{PQ} = 25\mathbf{I}$.

Without finding **Q**:

- (i) write down an expression for $\mathbf{P}^{-1}$ in terms of **Q**; (1)
- (ii) find the value of the constant k such that $(\mathbf{PQ})^{-1} = k\mathbf{I}$; (2)
- (iii) determine the numerical value of $\det \mathbf{Q}$ in the case when $a = 3$. (4)

(Total 12 marks)

Q17.

The matrices **A** and **B** are given by

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 3 & 8 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

The matrix $\mathbf{M} = \mathbf{A} - 2\mathbf{B}$.

- (a) Show that $\mathbf{M} = n \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$, where n is a positive integer. (2)

- (b) The matrix $\mathbf{M}$ represents a combination of an enlargement of scale factor p and a reflection in a line L . State the value of p and write down the equation of L . (2)

- (c) Show that

$$\mathbf{M}^2 = q\mathbf{I}$$

where q is an integer and $\mathbf{I}$ is the 2×2 identity matrix.

(2)
(Total 6 marks)

Q18.

- (a) The transformation T is defined by the matrix $\mathbf{A}$, where

$$\mathbf{A} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

- (i) Describe the transformation T geometrically. (2)

- (ii) Calculate the matrix product $\mathbf{A}^2$. (2)

- (iii) Explain briefly why the transformation T followed by T is the identity transformation. (1)

- (b) The matrix $\mathbf{B}$ is defined by

$$\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

- (i) Calculate $\mathbf{B}^2 - \mathbf{A}^2$. (3)

- (ii) Calculate $(\mathbf{B} + \mathbf{A})(\mathbf{B} - \mathbf{A})$. (3)
(Total 11 marks)

Q19.

The matrix $\mathbf{M}$ is defined by

$$\mathbf{M} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

(a) Find the matrix:

(i) $\mathbf{M}^2$;

(3)

(ii) $\mathbf{M}^4$.

(1)

(b) Describe fully the geometrical transformation represented by $\mathbf{M}$.

(2)

(c) Find the matrix $\mathbf{M}^{2006}$.

(3)

(Total 9 marks)

Q20.

The matrices $\mathbf{P}$ and $\mathbf{Q}$ are given by

$$\mathbf{P} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & t & -2 \\ 3 & 2 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{Q} = \begin{bmatrix} 1 & 1 & 1 \\ -7 & -1 & 5 \\ 11 & -1 & -7 \end{bmatrix}$$

where t is a real constant.

(a) Find the value of t for which $\mathbf{P}$ is singular.

(2)

(b) (i) Determine the matrix $\mathbf{R} = \mathbf{PQ}$, giving its elements in terms of t where appropriate.

(3)

(ii) Find the value of t for which $\mathbf{R} = k\mathbf{1}$, for some integer k .

(2)

(iii) Hence find the matrix $\mathbf{Q}^{-1}$.

(1)

(c) In the case when $t = -3$, describe the geometrical transformation with matrix $\mathbf{R}$.

(2)

(Total 10 marks)

Q21.

The matrices $\mathbf{A}$ and $\mathbf{B}$ are defined by

$$\mathbf{A} = \begin{bmatrix} 3 & 4 \\ 4 & 3 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix}$$

(a) Calculate the matrices:

(i) $\mathbf{A} + \mathbf{B}$;

(2)

(ii) $\mathbf{AB}$.

(2)

(b) Show that $\mathbf{A} + \mathbf{B} - \mathbf{AB} = k\mathbf{I}$, where k is an integer and $\mathbf{I}$ is the 2×2 identity matrix.

(2)

(Total 6 marks)



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